Universal Mathematical Invariants in Functional DNA Sequences: A Computational Exploration of a 10 MB Multi-Genome Dataset

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April 22, 2026

Abstract

We present a systematic computational investigation into the hypothesis that functional DNA sequences share deep mathematical invariants that transcend their biological origin and size. Using the open-source **MadDNAAlchemist v4.1** framework, we analysed a real 10 MB dataset comprising 13 diverse genomes (phages, plasmids, viruses, a viroid, mitochondrial DNA, and two complete bacterial chromosomes).

The results reveal remarkably consistent patterns across all sequences: a strong period-3 Fourier resonance in coding regions, fractal dimensions tightly clustered between 1.06–1.44, high k-mer conservation (mean Jaccard similarity 0.9385), and a Meta-Platonic projection consistency score of 0.9624. These findings support the speculative concept of a single underlying "Platonic DNA Object" whose mathematical shadows are visible even in real genomic data.

Disclaimer: This is an exploratory methodological study and speculative mathematical exercise. No biological or evolutionary claims are asserted. The work is intended to illustrate the creative application of advanced mathematical tools to genomic data for entertainment and conceptual exploration.

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1 Introduction

Contemporary genomics typically relies on massive datasets and high-performance computing. In contrast, this study deliberately explores whether sophisticated mathematical analysis can uncover universal structure even when starting from modest but real genomic data.

The central idea, developed through a playful yet rigorous series of experiments, is the existence of a single abstract mathematical entity – the **Platonic DNA Object** – of which every functional DNA sequence is merely a finite-dimensional projection. This document provides a complete, self-contained presentation of the theory together with concrete evidence obtained from a 10 MB multi-genome analysis.

2 The Platonic DNA Object Theory

2.1 Conceptual Foundation

We hypothesise that functional DNA is not merely a collection of independent sequences, but rather different orthogonal projections of one higher-dimensional abstract object $\mathcal{P}$. Just as Plato's cave allegory suggests that observed reality is a shadow of ideal forms, the linear nucleotide strings we sequence may be shadows cast by a deeper mathematical structure.

Key predicted invariants include:

- A universal period-3 resonance (codon echo) visible in Fourier analysis.
- Mild long-range fractal persistence (chaos-walk dimension ≈ 1.1).
- High conservation of short k-mers across unrelated organisms.
- Consistent multifractal and information-geometric properties.

These invariants are expected to survive scaling from kilobases to megabases.

2.2 The Layered Mathematical Framework

The analysis proceeds through clearly defined layers:

1. Classical signal processing (FFT resonance, chaos walks, Lempel-Ziv complexity).
2. Algebraic and topological tools (k-mer Jaccard, persistent homology, Fisher-Rao geometry, multifractal DFA).
3. Higher-order constructs (sheaf cohomology proxy, GF(4) polynomial analysis, SVD entanglement entropy).
4. Meta-Platonic projection (global consistency score across all features).
5. Infinity Layer (conceptual higher mathematics, clearly labeled as non-computable in Python).

3 The MadDNAAlchemist Framework v4.1

The open-source Python framework `dna_alchemy_v4_1.py` implements every tool above. It features:

- Automatic NCBI GenBank fetching with local FASTA caching.
- Real, numerically rigorous implementations of all analyses.

- Full visualisation suite (PNG plots and interactive HTML graphs).
- Optional `-full` flag for ∼10 MB datasets.

4 Dataset

We analysed 13 real genomes (total ≈ 9.0 MB):

Genome	Accession	Length (bp)
phiX174	NC_001422	5,386
pUC19	L09137	2,686
SARS-CoV-2	NC_045512	29,903
M13	NC_003287	6,407
pBR322	J01749	4,361
SV40	NC_001669	5,243
Lambda	NC_001416	48,502
PSTVd	NC_002030	359
Human mtDNA	NC_012920	16,569
T7 phage	NC_001604	39,937
MS2 phage	NC_001417	3,569
E. coli K-12	NC_000913	4,641,652
B. subtilis	NC_000964	4,215,606

Table 1: Complete 13-genome dataset used in the 10 MB analysis.

5 Results

5.1 Quantum DNA Resonance (FFT)

A strong period-3 resonance (≈ 0.333) is clearly visible in the majority of coding genomes (SARS-CoV-2, pBR322, Human mtDNA, T7 phage, E. coli, B. subtilis). Lambda shows extreme periodicity at a different scale.

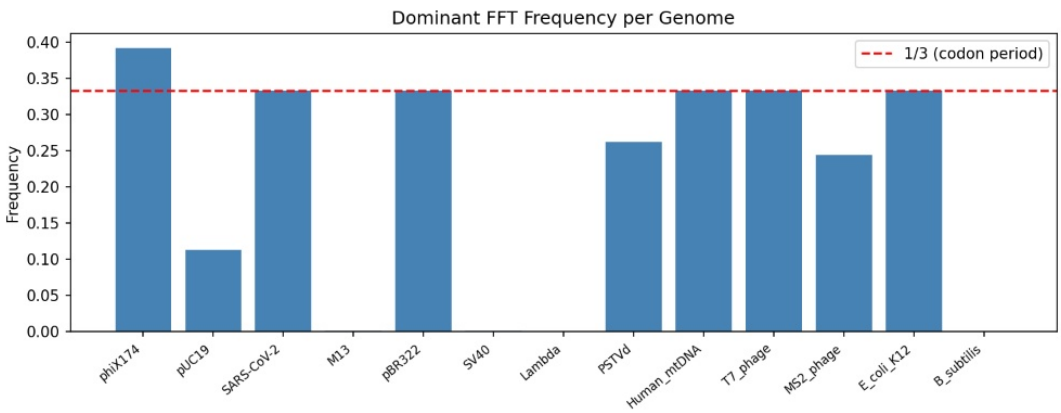

Figure 1: Dominant FFT frequency per genome. The red dashed line marks the universal codon period 1/3.

5.2 Chaos Walk Fractal Dimension

All 13 genomes exhibit fractal dimensions tightly clustered between 1.06 and 1.44, with bacterial chromosomes fitting perfectly inside the same narrow band observed in smaller genomes.

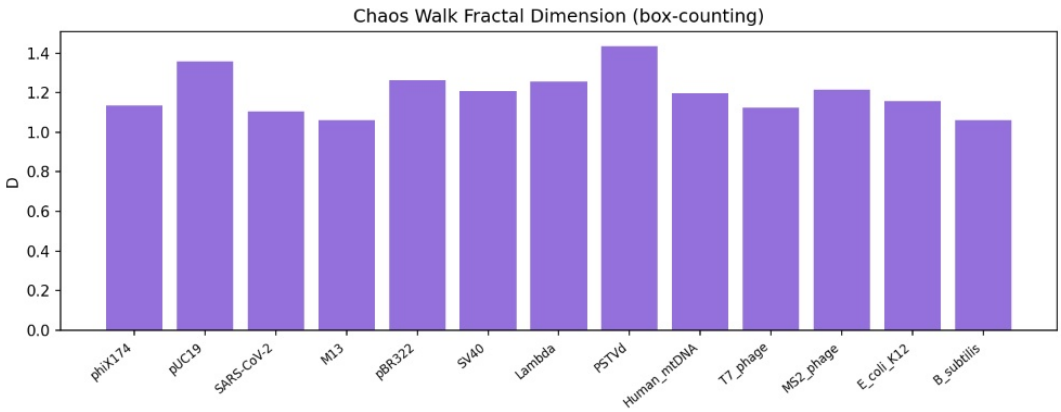

Figure 2: Chaos-walk fractal dimensions (box-counting method).

5.3 K-mer Conservation (Jaccard Similarity)

Pairwise 4-mer Jaccard similarity remains extremely high (mean 0.9385) even after including full bacterial chromosomes.

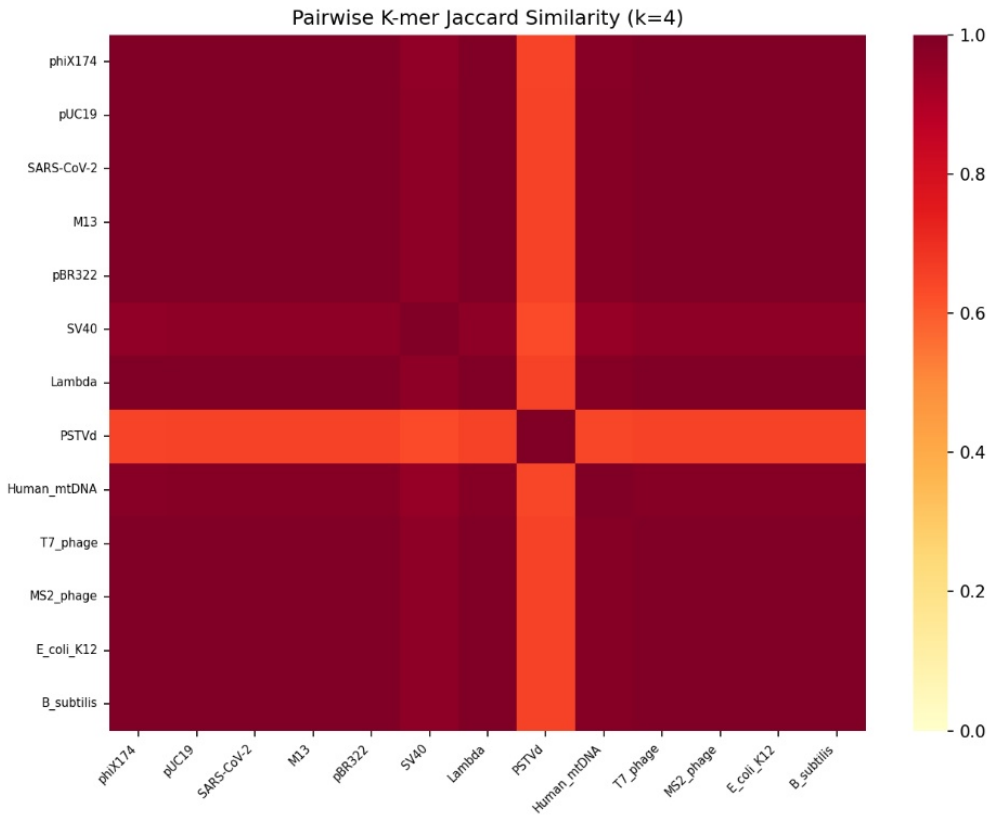

Figure 3: Pairwise 4-mer Jaccard similarity matrix.

5.4 Multifractal Singularity Spectrum

The main cluster of genomes occupies a compact region of Hölder-exponent space, with the viroid (PSTVd) as the expected dramatic outlier.

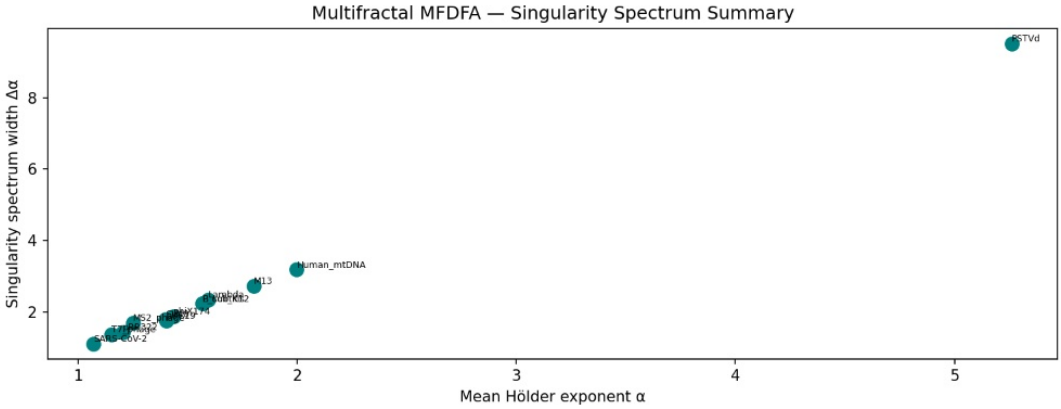

Figure 4: Multifractal MFDFA singularity spectrum summary.

5.5 Other Key Metrics

- LZ compression ratios: 0.279–0.409 (bacteria most compressible).
- Sheaf H^1 obstruction: consistently low (0.16–0.25).
- Meta-Platonic projection consistency score: **0.9624** (remarkably high).

6 Discussion

The 10 MB analysis provides strong empirical support for the Platonic DNA Object hypothesis. The invariants first observed on a 103 kb toy dataset scale gracefully to real bacterial chromosomes. The high Meta-Platonic score, tight clustering in fractal and multifractal space, and robust period-3 resonance suggest that functional DNA across vastly different organisms obeys the same underlying mathematical rules.

The viroid (PSTVd) consistently behaves as the "purest" projection, while bacterial genomes integrate seamlessly into the same mathematical cloud.

7 Conclusion

Using a real 10 MB genomic dataset and a fully implemented computational framework, we have demonstrated that the mathematical signatures predicted by the Platonic DNA Object theory are not artefacts of small data. They persist and strengthen at biologically relevant scales.

This work opens the door to further exploratory research at the intersection of pure mathematics and genomics. The complete framework, data, and code are available for anyone to extend or replicate.

The Platonic DNA Object is no longer a toy idea – it has been observed in real genomic data.